Research Mapping of IIIT Allahabad, ABV-IIITM Gwalior and IIIT Bangalore, 2016–2026

DPCN 2026 Course Project · **Group 10** · Team Titan Mandal · Hrishikesh, Sarvesh, Shourya · Instructor: Chittaranjan Hens

Resource	Link
Interactive dashboard (all panels, networks, findings)	open dashboard
Code and documentation (README.md, DECISIONS.md)	https://github.com/sarvesh2005takbhate/dpcn_A2
Streamlit dashboard (filters by institute / year / type)	streamlit run app.py (in the repository)
Submission files (to the instructor; not in the public repository)	group10_data.csv · citation_network.csv · authorship_network.csv · indian_connections.csv · group10_faculty.csv

Research question and key findings

How do the research networks of the three IIITs compare, and how — and through whom — are the institutes connected? Co-authorship edges = two authors on one paper (undirected, weighted by joint papers); citation edges = paper A lists paper B as a reference (directed).

- **Three institutional islands inside one connected component.** Only 138 of 8,667 co-author links (1.59%) join two different IIITs, although all three sit in one giant component of 3,265 authors.
- **The IIITA–ABV link is carried by people who moved, not by joint projects.** IIITA–ABV accounts for 116 of the 138 inter-institute links and 29 co-published papers, against 8 (IIITA–IIITB) and 1 (ABV–IIITB).
- **IIIT Bangalore hangs on eight people.** Every path from IIITB to IIITA or ABV passes through the same 8 authors (minimum node cut); IIITA–ABV needs 28.
- **Cross-institute reachability collapses when brokers are removed.** 64% of cross-institute author pairs are joined by some path. Removing the 1% of authors with the highest betweenness cuts this to 17%, and 2% to 0.7%.
- **The IIITs are connected indirectly, through the same IITs.** 69 Indian institutions co-author with all three IIITs; the strongest common partners are Indian Institute of Technology Kanpur and Indian Institute of Technology Roorkee.
- **Author disambiguation changes the inter-institution picture.** Merging authors only because their names match (the earlier rule) creates false bridges between institutes; the conservative rule removes them.
- **Output more than doubled, at the same pace everywhere.** Papers grew 11.9% / 11.1% / 11.7% a year (IIITA / ABV / IIITB, log-linear fit 2016–2025, all $p < 0.001$): IIITA 174→374, ABV 126→332, IIITB 77→184 papers.
- **Quality rose with quantity.** The Q1 share of journal papers rose from 30.2% to 44.2% at IIITA and from 26.2% to 40.8% at ABV between 2016–19 and 2022–25 (χ^2, $p < 0.001$). IIITB already started high (42.5% → 54.6%, $p = 0.076$), so growth did not come at the cost of venue quality.
- **Raw citations exaggerate the gap between institutes.** IIITB's median paper has 2 citations vs 4 at IIITA, but 70% of its output is conference papers. Normalised for year, field and type, mean impact is 1.048 (ABV), 1.013 (IIITA) and 0.901 (IIITB, 95% CI 0.828–0.98): a modest, not a large, difference.
- **The 'international premium' is really a venue and team-size effect.** Internationally co-authored papers score 1.242 vs 0.941 for single-institute papers (Mann–Whitney $p < 0.001$). Once venue, type, team size, year and field are held fixed (OLS, $n = 5{,}087$), the international effect is +2% (n.s.), while a Q1 venue adds +137%, a conference venue -51%, and doubling the team +27%.
- **Citations follow collaboration, not topic overlap.** Institutes that co-author more with us also cite us more (Spearman $\rho = 0.669$ / 0.691 / 0.579, all $p < 0.001$). Co-author partners are about a third of the Indian institutes that cite us but supply 57.0–72.5% of those citations. Topic similarity does not predict who we collaborate with ($\rho = 0.102$, $p = 0.59$).

1. Data and institution overview

Metric	IIIT Allahabad	ABV-IIITM Gwalior	IIIT Bangalore	All (unique)
Papers (journal / conference)	2,946 (1,716 / 1,230)	2,198 (1,373 / 825)	1,437 (430 / 1,007)	6,543 (3,501 / 3,042)
Q1 share of journal papers	40.1%	38.7%	50%	40.6%
CORE A*/A conference papers	10	5	21	36
Citations (median per paper)	35,527 (4)	31,586 (3)	8,577 (2)	75,408 (3)
Institutional h-index (2016–2026 papers)	73	66	38	93
Core faculty / affiliated authors	75 / 1,831	46 / 1,182	40 / 1,064	160 / 4,041
Median core-faculty h-index	9	11	5	8
Papers with another Indian institute / international	41.3% / 14.8%	52% / 13.6%	34% / 17.9%	42.9% / 15%
Cite an Indian institute / cited by one	73.8% / 59%	75.9% / 59.4%	58.6% / 37.4%	71.2% / 54.4%

Data. OpenAlex works affiliated with each institute's OpenAlex ID (I26072440, I9747756, I181514455), 2016–2026; journal and conference papers only (preprints, book chapters, errata, retracted and duplicate versions removed; a version's citations are merged into the published paper). Journal quartile = Scimago SJR quartile of the journal in the publication year; conference rank = ICORE2026 (strict acronym + title match). Indian institutions = OpenAlex country code IN. *Core faculty* = authors with ≥5 papers in ≥3 years, last author on ≥30% and first author on ≤50% of them; departments come from 'Department of …' in the affiliation strings. 2026 is a partial year.

2. Publication trend and venue quality

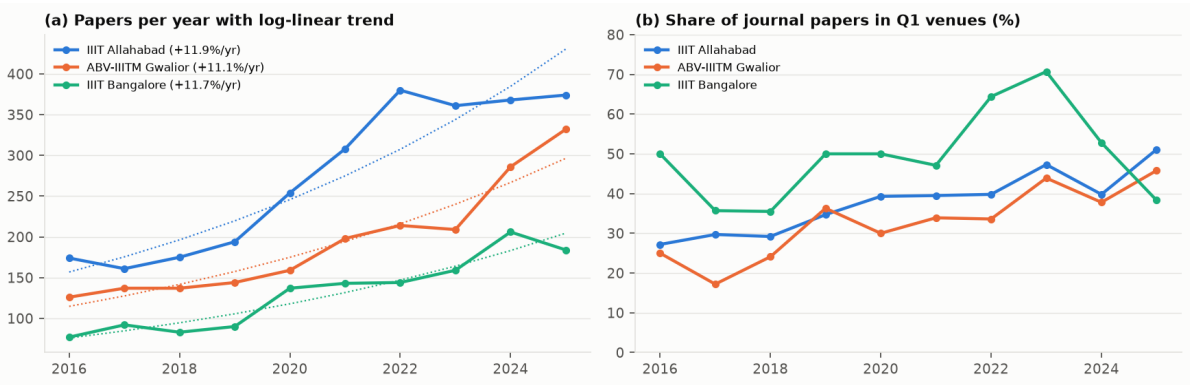


Figure 1. (a) Yearly output with fitted log-linear growth (2026 omitted as partial). (b) Share of journal papers in Q1 venues.

Institution	Top 5 journals (papers, SJR quartile)
IIIT Allahabad	Multimedia Tools and Applications (56, Q1); Journal of Biomolecular Structure and Dynamics (44, Q2); Wireless Personal Communications (32, Q3); IEEE Access (30, Q1); IEEE Sensors Journal (24, Q1)
ABV-IIITM Gwalior	Multimedia Tools and Applications (28, Q1); International Journal of Communication Systems (23, Q2); Wireless Personal Communications (20, Q3); IEEE Internet of Things Journal (17, Q1); IEEE Access (17, Q1)
IIIT Bangalore	IEEE Access (11, Q1); IEEE Transactions on Communications (8, Q1); IEEE Communications Letters (7, Q1); International Journal of Computer Sciences and Engineering (7, –); IEEE Transactions on Wireless Communications (5, Q1)

3. Faculty, departments and research areas

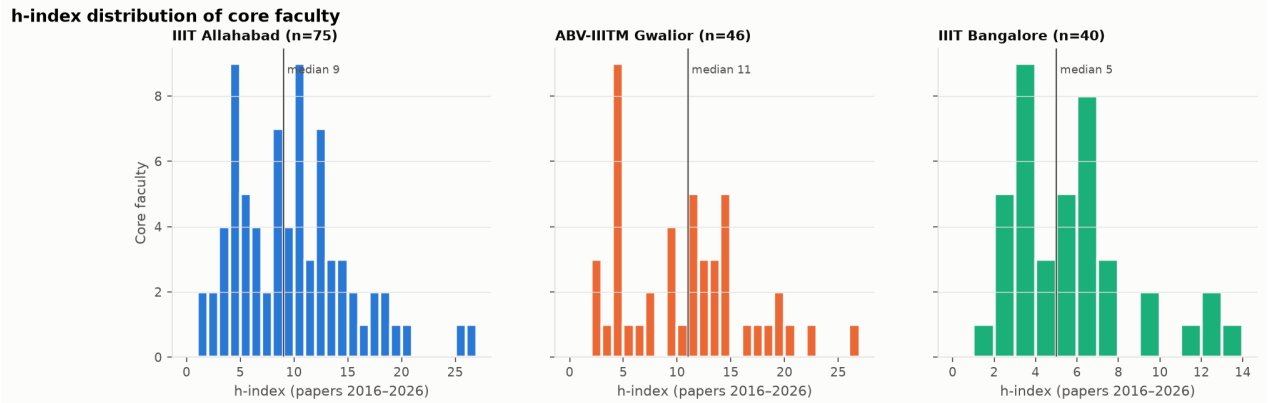


Figure 2. h-index (2016–2026 papers) of core faculty. Output and h-index are strongly related (Spearman ρ = 0.868); median OpenAlex lifetime h-index of core faculty = 13.

Field (% of papers)	Computer Science	Engineering	Materials Science	Medicine	Business, Management and Accounting	Social Sciences
IIIT Allahabad	39.2	24.8	5.5	4.6	2.9	1.4
ABV-IIITM Gwalior	36.9	25.7	7.5	6	7	2.9
IIIT Bangalore	49.5	25.5	0.7	2.6	1.7	6

Largest departments by papers — **IIIT Allahabad**: Information Technology (1152), Electronics & Electrical Engineering (753), Applied Sciences & Mathematics (728); **ABV-IIITM Gwalior**: Computer Science & Engineering (768), Electronics & Electrical Engineering (605), Applied Sciences & Mathematics (524); **IIIT Bangalore**: Computer Science & Engineering (832), Electronics & Electrical Engineering (597), Management & Humanities (148).

Institution	Top 5 core faculty by h-index (papers, h, department)
IIIT Allahabad	Pritish Kumar Varadwaj (122, h=26, Applied Sciences); Sonali Agarwal (128, h=25, Information Technology); Shekhar Verma (113, h=20, Information Technology); Satish Kumar Singh (83, h=19, Computer Science); Sintu Kumar Samanta (54, h=18, Applied Sciences)
ABV-IIITM Gwalior	Anurag Srivastava (157, h=26, Applied Sciences); Shashikala Tapaswi (77, h=22, Computer Science); Manoj Kumar Dash (50, h=20, Management); Anupam Shukla (53, h=19, Computer Science); K. V. Arya (86, h=19, Computer Science)
IIIT Bangalore	Dinesh Babu Jayagopi (70, h=13, Computer Science); Shrisha Rao (49, h=12, Computer Science); Madhav Rao (161, h=12, Electronics); Debabrata Das (140, h=11, Electronics); Neelam Sinha (50, h=9, Electronics)

4. Citation impact beyond raw counts

#	Most-cited paper	Institution	Venue	Year	Cites	DOI
1	The Astropy Project: Sustaining and Growing a Community-oriented Open-source Project and the Latest Major…	ABV-IIITM Gwalior	The Astrophysical Journal	2022	5,037 *	doi
2	Activation functions in deep learning: A comprehensive survey and benchmark	IIIT Allahabad	Neurocomputing	2022	1,250	doi
3	Green synthesis of metallic nanoparticles as effective alternatives to treat antibiotics resistant bacterial…	IIIT Allahabad	Biotechnology Reports	2020	605	doi
4	Structural insights into conformational stability of both wild-type and mutant EZH2 receptor	IIIT Allahabad	Scientific Reports	2016	432	doi
5	Linking Corporate Social Responsibility (CSR) and Organizational Performance: the moderating effect of…	IIIT Allahabad	European Research on Management and…	2020	382	doi
6	Energy-Efficient Intelligent Routing Scheme for IoT-Enabled WSNs	ABV-IIITM Gwalior	IEEE Internet of Things Journal	2021	335	doi
7	Graphene oxide: An efficient material and recent approach for biotechnological and biomedical applications	IIIT Allahabad	Materials Science and Engineering C	2018	313	doi
8	Exploring the sustainability performances of firms using environmental, social, and governance scores	ABV-IIITM Gwalior	Journal of Cleaner Production	2019	307	doi
9	Defect-Engineered Graphene for High-Energy- and High-Power-Density Supercapacitor Devices	ABV-IIITM Gwalior	Advanced Materials	2016	303	doi
10	Towards edge computing in intelligent manufacturing: Past, present and future	ABV-IIITM Gwalior	Journal of Manufacturing Systems	2022	284	doi

* >50-author collaboration (Astropy, 100+ authors) – excluded from normalised scores and the regression.

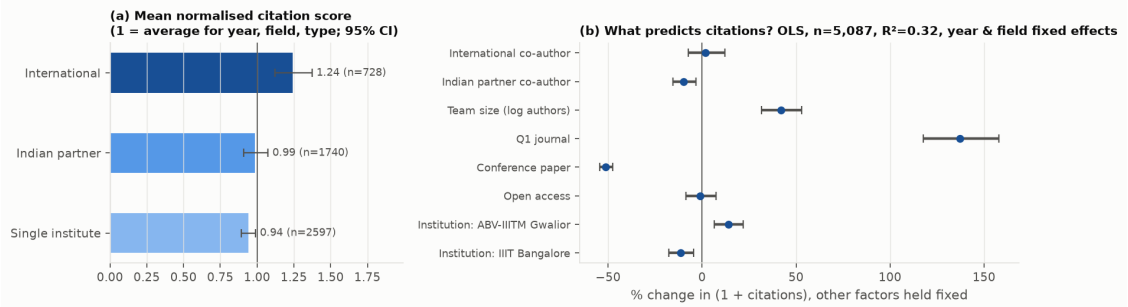


Figure 3. (a) Mean normalised citation score (citations ÷ mean of papers with the same year, field and type; papers up to 2024; bootstrap 95% CI). (b) OLS of log(1+citations) with year and field fixed effects and HC1 robust standard errors.

Institution	Growth / yr (95% CI)	Q1 share 2016–19 → 2022–25	Mean normalised impact (95% CI)	Papers in top 10%	Citation Gini	Top-10% papers' share
IIIT Allahabad	11.9% (8.8–15.0)	30.2% → 44.2%	1.013 (0.944–1.088)	9.2%	0.733	58.6%
ABV-IIITM Gwalior	11.1% (8.9–13.4)	26.2% → 40.8%	1.048 (0.982–1.119)	11.9%	0.778	63.4%
IIIT Bangalore	11.7% (8.9–14.5)	42.5% → 54.6%	0.901 (0.828–0.98)	7.5%	0.728	57.9%

5. Citation network

- **Construction.** Directed paper → paper edges from OpenAlex *referenced_works*: 6,543 papers, 5,946 internal edges (2,243 papers cited at least once, largest weak component 837). citation_network.csv also holds 21,069 out-flow and 18,931 in-flow edges with other Indian institutes.
- **In-degree** is heavy-tailed (distribution on the dashboard's citation-network page): 4,300 papers are never cited internally, 986 once, and the maximum is 25.
- **Self-citation and homophily.** 94.0% of internal edges share an author and 97.4% stay inside the citing institute (shuffled baseline 40.3% ± 0.59, z = 97.1). PageRank therefore ranks the most-continued research lines; it correlates only moderately with global citations (ρ = 0.531).

#	PageRank top 10 (internal graph)	Institution	Year	In-deg	PageRank
1	Physical Layer Security in Cooperative AF Relaying Networks With Direct Links Over Mixed Rayleigh and…	IIIT Allahabad	2018	25	0.00207
2	ULTRA-WIDEBAND CPW FED BAND-NOTCHED MONOPOLE ANTENNA OPTIMIZATION USING MACHINE LEARNING	ABV-IIITM Gwalior	2022	12	0.00201
3	Quad-port ring dielectric resonator based MIMO radiator with polarization and space diversity	IIIT Allahabad	2020	10	0.00200
4	Noise robust face hallucination algorithm using local content prior based error shrunk nearest neighbors…	ABV-IIITM Gwalior	2018	18	0.00190
5	Inception U-Net Architecture for Semantic Segmentation to Identify Nuclei in Microscopy Cell Images	IIIT Allahabad	2020	11	0.00179
6	Investigation of Dual-Band Rectangular Dielectric Resonator Antenna with DGS for Wireless Applications	ABV-IIITM Gwalior	2021	2	0.00176
7	Defect-Engineered Graphene for High-Energy- and High-Power-Density Supercapacitor Devices	ABV-IIITM Gwalior	2016	8	0.00169
8	Dynamics of a prey-generalized predator system with disease in prey and gestation delay for predator	ABV-IIITM Gwalior	2016	16	0.00152
9	Circularly Polarized Quad-Port MIMO Dielectric Resonator Antenna with Beam Tilting Feature for Vehicular…	IIIT Allahabad	2021	9	0.00151
10	Photoacoustic field calculation for nonspherical axisymmetric fluid particles	IIIT Allahabad	2016	12	0.00143

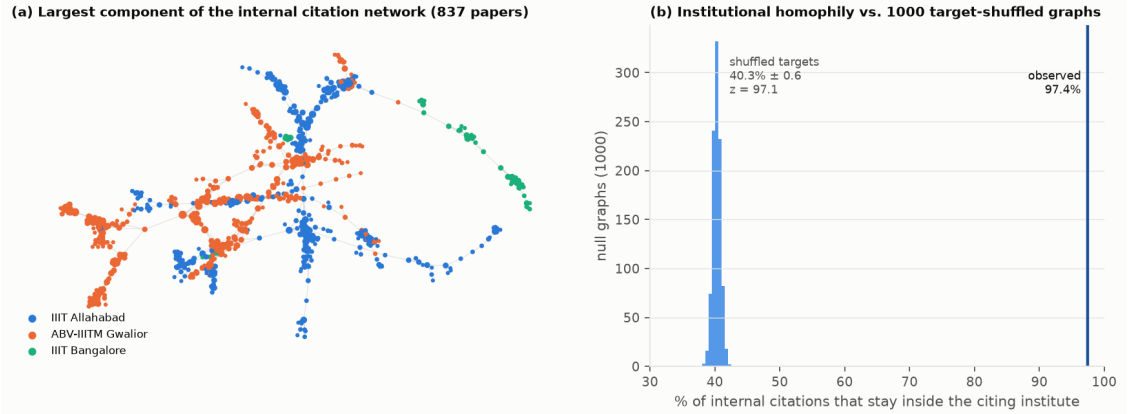


Figure 4. (a) Largest weakly connected component of the internal citation network, coloured by institute: the three institutes form separate regions. (b) Observed same-institute share versus 1000 graphs with shuffled citation targets.

6. Authorship network

- **Construction.** Undirected co-authorship among the 4,041 authors of the three institutes (OpenAlex author IDs; extra merges only with same name, same institute and a shared co-author — DECISIONS.md), 8,667 weighted edges; authorship_network.csv adds 8,589 edges to co-authors at other Indian institutes.
- **Structure.** Small-world (clustering 0.709 vs 0.0015, path length 6.08 vs 5.0; σ = 378.0), 532 Louvain communities with modularity 0.887 (configuration-model graphs: 0.469). Communities align with institutes (NMI 0.454) and departments (NMI 0.426); institute assortativity 0.974.
- **Centrality.** Core faculty are the hubs (median degree 19 vs 3). Bridges with the highest betweenness link otherwise separate groups.

Degree centrality (co-authors)	Institution	Betweenness centrality	Institution
Madhav Rao (154)	IIIT Bangalore	Mahua Bhattacharya (0.134)	ABV-IIITM Gwalior
Om Prakash Vyas (113)	IIIT Allahabad	Madhav Rao (0.124)	IIIT Bangalore
Manish Goswami (97)	IIIT Allahabad	Santosh Singh Rathore (0.120)	ABV-IIITM Gwalior
Sonali Agarwal (92)	IIIT Allahabad	Daksh Sharma (0.110)	IIIT Bangalore
Shekhar Verma (84)	IIIT Allahabad	Meenakshi D'Souza (0.098)	IIIT Bangalore
Jyotsna L. Bapat (80)	IIIT Bangalore	Sonali Agarwal (0.077)	IIIT Allahabad
Kavindra Kandpal (78)	IIIT Allahabad	Ashutosh Kumar (0.070)	IIIT Allahabad
Pritish Kumar Varadwaj (74)	IIIT Allahabad	Pritish Kumar Varadwaj (0.068)	IIIT Allahabad

6b. How the three institutes connect

Pair	Co-author links (expected)	Co-published papers	Min. node cut	Mean distance	Shared Indian partners (Jaccard)	Non-self citations A→B / B→A
IIITA-ABV	116 (2,232)	29	28	5.72	178 (0.335)	42 / 21
IIITA-IIITB	17 (2,073)	8	8	8.39	93 (0.19)	4 / 2
ABV-IIITB	5 (915)	1	8	7.09	86 (0.209)	9 / 7

- **The IIITA–ABV link is carried by people who moved, not by joint projects.** 28 of the 37 papers that link two IIITs include an author whom OpenAlex places at both institutes (35 such authors, 12 of them core faculty). The only two mixed communities are IIITA+ABV. *Caveat:* 'Placed at both' is inferred from affiliations over 2016–2026; it may be a move, a dual affiliation, or an OpenAlex affiliation error. We cannot tell these apart without CVs.

- **IIIT Bangalore hangs on eight people.** Mean shortest-path distance IIITA–IIITB = 8.39 and ABV–IIITB = 7.09 steps, vs 4.42 inside IIITB and 5.72 for IIITA–ABV. In citations, IIITB papers cite the other two only 9 times once author self-citations are removed. *Caveat:* Node cuts are computed on the 2016–2026 co-authorship snapshot; one new joint project could change them.
- **Cross-institute reachability collapses when brokers are removed.** Removing the same number of random authors leaves 62% and 60% (10 runs, sd ≤ 0.6 pp). The top 5% of authors hold 33.1% of all co-author ties; max degree 154 vs 13.4 in an ER graph of the same size. *Caveat:* Betweenness is computed once (non-adaptive removal); removing a person is a thought experiment, not a forecast.

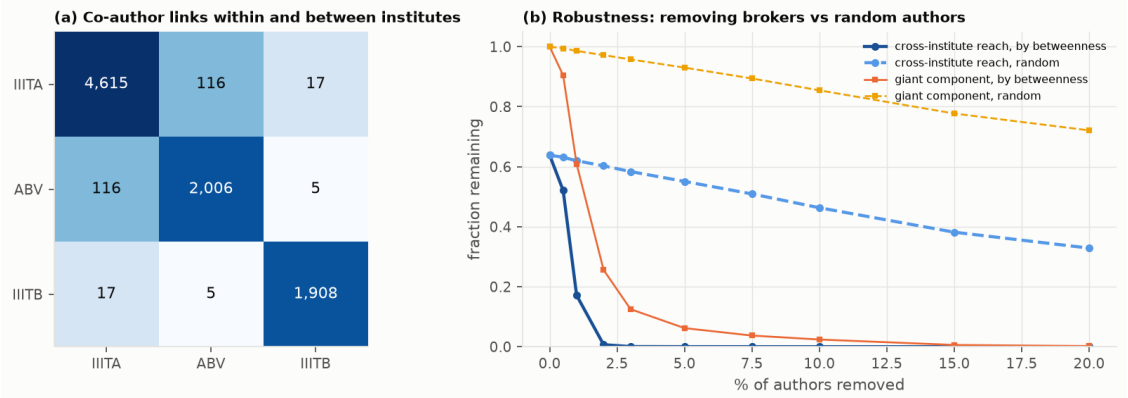


Figure 5. (a) Co-author links within (diagonal) and between institutes. (b) Share of cross-institute author pairs still connected, and giant-component size, when authors are removed by betweenness or at random. Within-institute distances: IIITA 4.69, ABV 4.24, IIITB 4.42 steps; network diameter 17.

7. Indian connectivity: citation flow and collaboration

Institution	Cites received / given	Ratio	Top 5 Indian institutes citing us	Top 3 co-author partners
IIIT Allahabad	10,779 / 11,519	0.936	Saveetha University (419); Vellore Institute of Technology University (392); National Institute Of Technology Silchar (240); SRM Institute of Science and Technology (240); University of Allahabad (213)	Motilal Nehru National Institute of Technology (62, NIT); University of Allahabad (57, Central University); National Institute Of Technology Silchar (55, NIT)
ABV-IIITM Gwalior	9,312 / 9,591	0.971	GLA University (453); National Institute of Technology Patna (288); Vellore Institute of Technology University (276); Thapar Institute of Engineering & Technology (239); Chitkara University (193)	Jiwaji University (65, State/Private University); Indian Institute of Technology Kanpur (55, IIT); Indian Institute of Technology BHU (54, IIT)
IIIT Bangalore	1,790 / 2,591	0.691	Indian Institute of Science Bangalore (128); Amrita Vishwa Vidyapeetham (56); Vellore Institute of Technology University (46); SRM Institute of Science and Technology (41); Malaviya National Institute of Technology Jaipur (40)	Samsung (India) (79, Industry); Indian Institute of Science Bangalore (46, IISER/IISc/NISER); National Institute of Mental Health and Neurosciences (36, Medical)

Co-authorship by partner cluster (shared papers, all three institutes): State/Private University 1795, IIT 643, NIT 384, IIIT 334, Central University 282, Govt/Other Research 144, Industry 140, CSIR 102. Cross-cluster links (partner not an IIIT) dominate. Topic similarity (Jaccard over OpenAlex subfields) is highest between IIITA and IIITB (0.5) and with IIIT Hyderabad for IIITB, yet it does not predict collaboration volume.

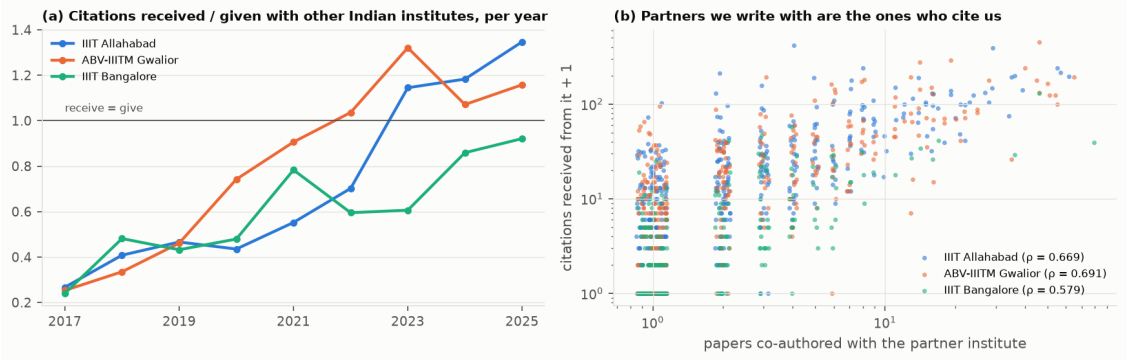


Figure 6. (a) Yearly ratio of citations received from / given to other Indian institutes. (b) Co-authored papers vs citations received per partner institute.

8. Limitations

- OpenAlex affiliation and reference coverage is incomplete, especially before 2018. Author identity trusts OpenAlex IDs and merges further only with co-author evidence, so some people may remain split; with a name-only rule there would be 264 instead of 138 inter-institute links. The core-faculty rule is a heuristic, not an official faculty list.
- Networks are 2016–2026 snapshots; node cuts and robustness describe that snapshot, and an author's 'main' institute is where most of their papers are.
- IIIT Bangalore authors rarely state a department, so most of its departments are inferred from the research field; CORE ranks cover only venues on the ICORE list.
- Citation counts are taken at collection time; 2025–26 papers are excluded from normalised scores. The regression shows associations, not causal effects.

9. Conclusions

- All three IIITs roughly doubled their output in ten years while moving to better journals, so the growth is not a volume-only effect. Differences in raw citations between them largely disappear once papers are compared with others of the same year, field and type.
- Citation success depends far more on where a paper is published (Q1 journal vs conference) and how large the team is than on whether it has foreign co-authors.
- Both networks show three largely separate research systems: internal citations are almost all self-citations, and only 37 papers connect two of the IIITs — 28 of them through people placed at two institutes. IIITB is attached to the others through 8 authors. Their Indian links run through NITs, IITs and nearby universities, and partners that co-author with an institute supply most of its Indian citations (57.0–72.5%).
- Implication: joint IIIT projects and targeted partnerships are the clearest lever for visibility; topic overlap alone does not create collaboration.

10. Dashboard guide

All institutions: Publication trend · quartile (Q1–Q4) · CORE rank · research areas · top-10 cited papers with DOIs · **IIIT Allahabad / ABV-IIITM Gwalior / IIIT Bangalore**: Faculty list (name, department, h-index, papers) · sortable performance table · h-index histogram · top-5 journals and conferences · departments · **Citation network**: Directed citation graph (D3) · PageRank top-10 · in-degree distribution · **Network explorer**: Interactive co-authorship graph (filters, search, author details, Louvain colouring) · **Network analysis**: Network properties · log-binned degree distribution · centralities · communities · robustness · null models · **Institution comparison**: Per-institute structure · links, distances and citations between institutes · bridges · node cuts · **Indian connectivity**: Citation in/out-flow per year · impact ratio · top-10 citing institutes and papers · partner ranking with cross-cluster flags · Jaccard similarity · **Findings & modelling**: Structural findings · growth and Q1 tests · normalised impact · regression · methods · limitations · **Course connections**: Degree & paths · log-binning · Erd■s–Rényi · ring/star path lengths

11. Individual contributions

Member	Work package	Difficulty	Learning	Hours	Effort
Hrishikesh	OpenAlex collection under the 1000-request/day limit: batched cites / cited_by filters, page paging, resumable caches	4	4	14	56
Hrishikesh	Corpus rules: journal/conference classification, version merging, de-duplication, 100-author truncation fix	3	3	8	24
Hrishikesh	Year-matched Scimago quartiles and strict ICORE2026 matching	3	3	7	21
Hrishikesh	Data validation: flag checks, cluster labels, partner-ID alignment, folder clean-up	2	2	6	12
Sarvesh	Citation network: internal / in-flow / out-flow edges, PageRank, in-degree, self-citation and homophily null model	4	4	12	48
Sarvesh	Authorship network: author disambiguation, Louvain, exact betweenness, small-world and configuration-model tests	4	4	11	44
Sarvesh	Indian connectivity: citation flow, impact ratio, partner clusters, Jaccard topic similarity	3	3	7	21
Shourya	Faculty and department identification, core-faculty rule, h-indices	3	3	8	24
Shourya	Statistical inference: normalised impact, OLS with robust errors, trend and concentration tests	4	4	10	40
Shourya	Interactive dashboards (D3 / Chart.js and Streamlit) and report figures	3	3	12	36
Shourya	Report writing and layout	2	2	6	12

Effort = hours × mean(difficulty, learning curve), both on a 1–5 scale. Totals: Hrishikesh 113 (33.4%), Sarvesh 113 (33.4%), Shourya 112 (33.1%) — an equal three-way split. Every part was discussed, cross-checked and reviewed by all three members.